

eDrishti: Revolutionising STEM Education for the Visually Impaired

SYNOPSIS

In response to the staggering educational challenges faced by visually impaired individuals in science and mathematics, eDrishti presents a groundbreaking solution. With only 5% of visually impaired people globally and a mere 1% in India being literate, the lack of accessible study materials exacerbates the issue, particularly in government schools.

Inspired by legal cases and personal anecdotes highlighting the struggles of visually impaired students, eDrishti aims to transcend these barriers by making science and mathematics education accessible beyond the 8th standard. Our innovative software-embedded system comprises a mobile application and connected hardware, designed to revolutionize STEM learning for the visually impaired.

KEY FEATURES

1. Software Component

The mobile application employs advanced voice explanation and visualization techniques to elucidate STEM concepts. Guided by blind tutors, users can create customized learning modules tailored to their needs. With TalkBack and voice control features, accessibility is prioritised for seamless navigation.

2. Hardware Component

Our unique hardware system utilizes pins that rise and fall based on geometrical shapes traced on the software. This tactile approach enables users to engage with abstract concepts physically, fostering deeper understanding and retention.

INNOVATION

- Comprehensive STEM education software embedded system
- 24/7 support from blind tutors
- Affordable and scalable solution
- Integration of real-time geometry and science practice via hardware and software synergy

PROTOTYPE & DEVELOPMENT

The current prototype encompasses both software and hardware components, with the software offering access to study materials up to the 8th standard. Notably, the mobile application is fully compatible with screen reader technology, ensuring inclusivity. Furthermore, the hardware, constructed from solenoid door locks, enables tactile exploration of geometrical figures via Bluetooth connectivity.

DEMONSTRATION

- [Link](#)

USER TESTIMONIALS

- [Link 1](#)
- [Link 2](#)

KICKSTART 6-MONTH PLAN

- To finalize the development of MVP hybrid application and hardware
- To conduct extensive user testing and gather feedback from visually impaired individuals and educators
- To refine user interface and accessibility features based on feedback
- To forge partnerships with NGOs, blind schools, and government agencies for distribution and promotion
- To launch a pilot program in select regions to evaluate usability and impact
- To iterate based on pilot program results in preparation for full-scale market launch

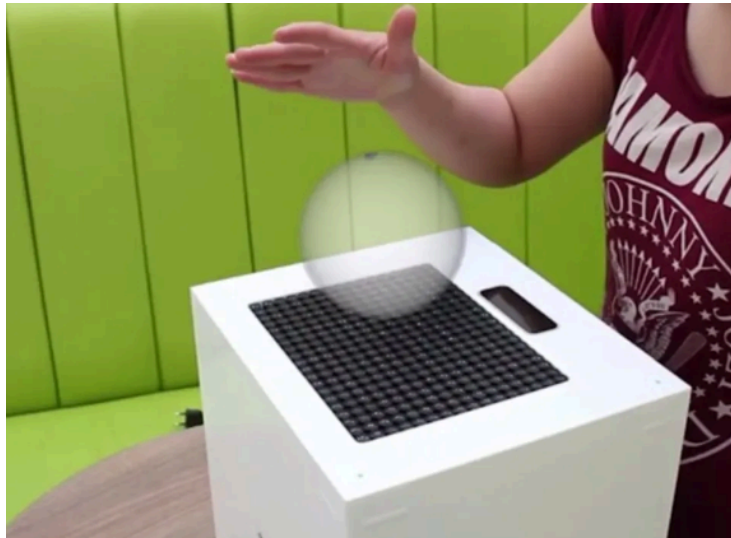
ACHIEVEMENTS

- Collaboration with prominent NGOs, Microsoft, and VOSAP
- Recognition from Microsoft CEO Satya Nadella, with a [PR event](#) highlighting eDrishti
- Award-winning [research paper](#), titled “Best Research Paper 2023”
- Patent publication: 202311009452
- Recipient of Microsoft Imagine Cup National Award 2023
- Selected among the Top 100 Social Impact Projects under the Google Solution Challenge 2023
- Selection in Tech for Change Program Cohort-3 by Nasscom Foundation and Cienna
- Winner in UNICEF subsidiary, VOSAP, international hackathon
- Over 20+ awards garnered in various hackathons

FUTURE SCOPE

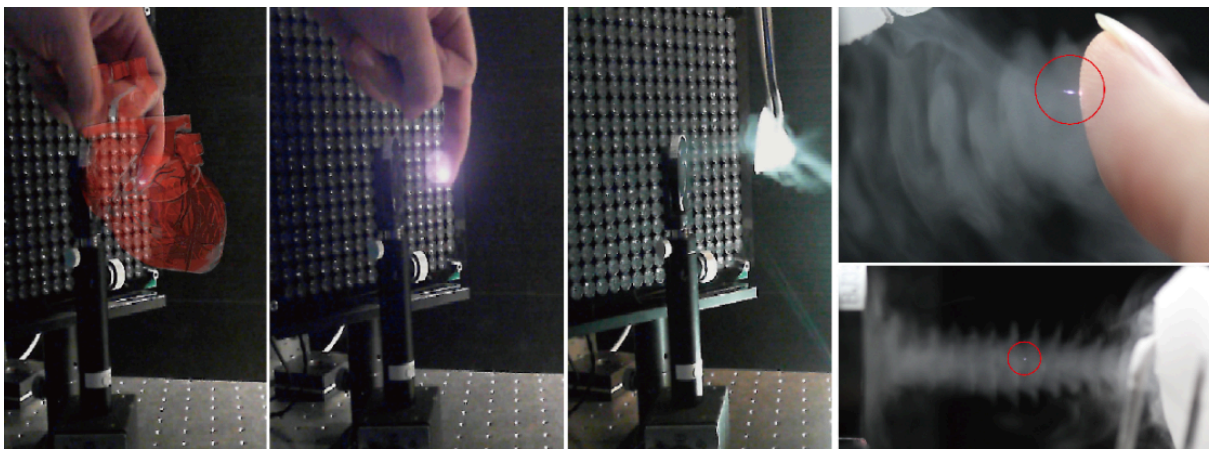
- **Haptic system development**
 - Utilizes ultrasonic waves to produce tactile sensations in mid-air.
 - Employs ultrasonic transducers emitting high-frequency sound waves (>20 kHz).
 - Driven by an electronic circuit generating ultrasonic signals.
 - Controls spatial distribution via phase and amplitude modulation of emitted waves.
 - Creates localized high-pressure and low-pressure zones in mid-air.
 - The intersection of ultrasonic waves generates acoustic radiation force.
 - Utilized to exert pressure on objects or particles within the intersecting region.

→ Ultrasonic visualisation of 3d objects



Source - [link](#)

- Ultrasonic sensors integrated with a Braille display offer real-time feedback on distance and form of 3D objects, enhancing spatial awareness for the visually impaired.
- Translation of scanned data into 3D maps improves navigation and independence.
- Customizable settings and compatibility with navigation systems enhance flexibility and usability.
- Additional feedback mechanisms like vibration or auditory signals provide extra information for users.



Source - DOI:10.1145/2858036.2858489

→ Sound response system

- Enhance accessibility by integrating sound response into a tactile system, providing both tactile and aural feedback.
- Sensors will monitor surroundings, converting data into Braille and auditory cues.

- Users navigate with traditional Braille input methods, receiving additional auditory feedback.
- This multi-modal approach accommodates diverse user preferences, improving usability for the visually impaired.

COMMERCIALIZATION

→ Market Research and Validation

- Conduct thorough market research to comprehend visually impaired students' and educators' needs.
- Validate demand for technology-driven solutions (e.g., sound response systems, ultrasonic visualization of 3D objects, haptic systems) through surveys, focus groups, and pilot programs.

→ Business Model Development

- Forge partnerships with educational institutions, special education associations, assistive technology providers, and government agencies.
- Collaborate with content developers and educational technology companies to integrate the technology into existing curriculum frameworks.

→ Product Development and Iteration

- Current Status: Proof of Concept (POC) Software and Hardware Ready.
- Next 6 months: Focus on developing Minimum Viable Product (MVP) software and hardware, incorporating feedback from target groups and iterating the product.

→ Marketing and Promotion

- Develop a targeted marketing strategy to raise awareness about the technology among educators, administrators, and policymakers.
- Highlight the technology's benefits in improving access to education, enhancing learning outcomes, and fostering inclusivity in the classroom.

→ Distribution and Sales

- Establish efficient distribution channels to make the technology accessible to educational institutions and students, especially in India.
- Partner with assistive technology distributors, educational resellers, and online platforms to facilitate sales and distribution.

→ Scaling and Expansion

- Strategically scale technology adoption by expanding into new markets and educational sectors.
- Invest in infrastructure, support services, and professional development to support widespread technology adoption in schools and educational programs globally.